MILITARY SPECIFICATION

SCALES, GRAPHICAL FIRING: XM75E1, XM76E1, AND XM77E1

1. SCOPE

1.1 This specification covers performance requirements, related quality assurance provisions, and packaging data for three graphical firing scales, each consisting principally of a rule, a slide, and an indicator, for calculating firing data corresponding to the applicable ammunition (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-12937	Scales and Tables, Graphical Firing: Scales and Tables, Graphical Site; Packaging of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

F10533330	Scale, Graphical Firing: XM75E1
F10533335	Scale, Graphical Firing: XM76E1
F10533340	Scale, Graphical Firing: XM77E1

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The scales shall be manufactured in accordance with the applicable drawings listed in Section 2 and all drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.3 Environmental service conditions.

3.3.1 Storage temperatures.- The scales shall show no evidence of physical failure after having been exposed to, and thermally stabilized at, ambient temperatures of -80° and +160° Fahrenheit (F).

3.3.2 Standard ambient temperature.- Subsequent to the requirement of 3.3.1 the scales shall meet the requirements of 3.4 to 3.8 inclusive at a standard ambient temperature between +60° and +90°F.

3.3.3 Extreme operating temperatures.- The scales shall meet the requirements of 3.8 while exposed to, and thermally stabilized at, ambient temperatures of -40° and +150°F.

3.3.4 Humidity.- The scale shall meet the Procedure I humidity requirements of MIL-F-13926. The scale shall then be thermally stabilized after return to a standard ambient temperature between +60° and +90°F., and shall meet the requirements of 3.4.

3.4 Post-environmental.- Subsequent to the requirements of 3.3.1 and 3.3.4, the scales shall show no color changes, distortion, warpage, breaks, stickiness, cracks, mottling, or separation of bonded materials, and the scales shall meet the requirements of 3.5 to 3.8 inclusive.

3.5 Printing and surface quality.- Letters and numbers shall be in accordance with the applicable drawings. They shall be sharp, evenly colored, without skips or breaks, and shall not touch each other or any line of the graduated scales. The printing surface shall be free from stains and other imperfections.

3.6 Graduations.- Graduations shall be accurately located in accordance with dimensions and related tolerances shown on the appropriate drawings. They shall be sharp, evenly colored, without skips or breaks.

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3.7 Accuracy.

3.7.1 "C" and "D" scales.- With the first graduation of the "C" scale in coincidence with the first graduation of the "D" scale, each succeeding graduation on the "C" scale shall be in alignment with its respective graduation on the "D" scale within 0.004 inch.

3.7.1.1 A 2/3 ratio between graduations on the "C" and "D" scales shall be read throughout the scale within the tolerance indicated in 3.7.1 when the first graduation line on the "C" scale is set in coincidence with the "15" graduation on the "D" scale.

3.7.2 Site and angle of site readings.- Readings for site and angle of site shall be accurate within the tolerances specified in Table I for the vertical intervals and ranges listed.

TABLE I

SCALES, GRAPHICAL FIRING: XM75E1, XM76E1, AND XM77E1

ANGLE OF SITE COMPUTATIONS		
VERTICAL INTERVAL (Meters) (Set hairline of indicator on "D" Scale)	RANGE (Meters) (Indicate under hairline)	ANGLE OF SITE (Mils) (Read on "D" Scale opposite M gage point)
-20	1500 (on "C" scale)	-13.58 ± 0.01
+30	4000 (on "C" scale)	+ 7.64 ± 0.01
+190	7000 (on "C" scale)	+27.65 ± 0.02

SITE COMPUTATIONS			
GRAPHICAL FIRING SCALE	VERTICAL INTERVAL (Meters) (Set hairline of indicator on "D" scale)	SITE RANGE (Meters) (Indicate under hairline)	SITE (Mils) (Read on "D" scale opposite M gage point)
XM75E1	+20	1200 (on "TAG" scale)	+18.61 ± 0.02
	-18	1400 (on "TBG" scale)	-14.64 ± 0.01
	+ 6	800 (on "TAG" scale)	+ 7.76 ± 0.01
	-30	1000 (on "TBG" scale)	-31.57 ± 0.03

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TABLE I (Continued)

SCALES, GRAPHICAL FIRING: XM75E1, XM76E1, AND XM77E1

SITE COMPUTATIONS			
GRAPHICAL FIRING SCALE	VERTICAL INTERVAL (Meters) (Set hairline of indicator on "D" scale)	SITE RANGE (Meters) (Indicate under hairline)	SITE (Mils) (Read on "D" scale opposite M gage point)
XM76E1	+20	1200 (on "TAG" scale)	+17.89 ± 0.02
	-18	1400 (on "TBG" scale)	-12.90 ± 0.01
	+ 6	800 (on "TAG" scale)	+ 7.70 ± 0.01
	+30	1000 (on "TAG" scale)	+31.03 ± 0.03
XM77E1	+25	1800 (on "TAG" scale)	+14.24 ± 0.01
	-14	2100 (on "TBG" scale)	- 6.63 ± 0.01
	+35	3900 (on "TAG" scale)	+ 2.625 ± 0.002
	-110	3500 (on "TBG" scale)	-42.1 ± 0.03

3.7.3 Superelevation readings.- Readings of the superelevation scale shall be accurate within the tolerances specified in Table II for the ranges listed.

TABLE II

SCALES, GRAPHICAL FIRING: XM75E1, XM76E1, AND XM77E1

GRAPHICAL FIRING SCALE	RANGE (Meters)	SUPERELEVATION (Mils)
XM75E1	1000	224.7 ± 0.1
	1400	347.2 ± 0.2
	1800	524.3 ± 0.3
XM76E1	1200	249.3 ± 0.1
	1600	356.3 ± 0.2
	2100	560.1 ± 0.3
XM77E1	2100	217.0 ± 0.1
	2900	333.6 ± 0.2
	3800	549.6 ± 0.3

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3.8 Operability.- The slide and indicator shall move freely in relation to the rule when the scale is exposed to, and thermally stabilized at, the standard ambient temperature of 3.3.2 and the extreme operating temperatures of 3.3.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plans specified in Table III titled "Classification of Defects". Examinations and tests for packaging, packing and marking shall be in accordance with MIL-S-12937 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

		REQUIREMENTS	TEST PROCEDURES
CRITICAL: NONE			
MAJOR: AQL 1.0 percent defective			
101.	Printing and surface quality	3.5	4.5.6
102.	Graduations	3.6	4.5.7
103.	Accuracy - "C" and "D" scales	3.7.1	4.5.8, 4.5.8.1
104.	Accuracy - site and angle of site readings	3.7.2	4.5.9
105.	Accuracy - superelevation readings	3.7.3	4.5.10
106.	Operability - slide and indicator (60° to 90°F)	3.8	4.5.11
MINOR: NONE			

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4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Standard sampling.

4.3.1 General.- One of each scale shall be selected at random by a Government representative as a standard sample from each 100 produced. The sample shall meet the requirements and tests in Table IV.

TABLE IV

CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings - 4.5.1
302. General specification	3.2	MIL-F-13926 - visual

4.3.2 Environmental.- Three of each scale shall be selected at random by a Government representative as a standard sample from each 50 produced or from each month's production, whichever occurs first. The samples shall have met the requirements and tests in Table III and shall then meet the requirements and tests in Table V.

TABLE V

CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURES
303. Storage temperatures	3.3.1	4.5.2
304. Operating temperatures - slide and indicator (-40° and +150°F)	3.3.3	4.5.3
305. Humidity	3.3.4	4.5.4

4.3.3 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied

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by the contractor in accordance with the requirements of 4.4.

4.4.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5 Test methods and procedures.

4.5.1 Fabrication.- The graphical firing scales specified shall be subjected to a visual, tactile, and dimensional inspection for conformance to the applicable drawings. Standard measuring equipment shall be used in performing dimensional inspection. The graphical firing scales shall conform to the requirement of 3.1.

4.5.2 Storage temperatures.- The scales shall be exposed to the temperatures specified in 3.3.1, utilizing testing equipment conforming to the "Test Facilities" requirements of MIL-F-13926. The equipment shall be capable of maintaining the specified temperatures until thermal stabilization is reached. The temperatures shall be changed gradually to avoid thermal shock. The scales shall lay on a flat surface in the test chamber, and the flat surface shall support the entire length and width of each scale. The scales shall be exposed to each specified temperature for a period of 5 hours. After being exposed to the above conditions, and upon return to the standard ambient temperature of 3.3.2, the scales shall be subjected to the test in 4.5.5 and shall conform to the requirements specified.

4.5.3 Extreme operating temperatures.- The scales shall be exposed to the temperatures specified in 3.3.3, utilizing testing equipment conforming with the "Test Facilities" requirements of MIL-F-13926. The equipment shall be capable of maintaining the specified temperatures until thermal stabilization is reached. The temperatures shall be changed gradually to avoid thermal shock. When thermal stabilization is reached at each temperature extreme, the scales shall be tested in accordance with 4.5.11 to determine operability in compliance with 3.8 as required by 3.3.3.

4.5.4 Humidity.- This test shall be conducted in accordance with Procedure I of the humidity test in MIL-F-13926. Upon completion of this test, and after the scales have been thoroughly dried, the scales shall be subjected to the tests in 4.5.5 and shall conform to the requirements referenced therein, as required by 3.3.4.

4.5.5 Post-environmental.- Subsequent to the tests specified in 4.5.2 and 4.5.4, the scales shall be subjected to a visual and tactile examination for damage or defects as specified in 3.4. The scales shall then be subjected to the tests specified in 4.5.6 to 4.5.11 inclusive, and shall conform to the requirements referenced therein, as required by 3.4.

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4.5.6 Printing and surface quality.- The printing and surface quality of the scale shall be visually inspected with the aid of a magnifier of at least 5 power. The printing and surface quality of the scale shall conform with the requirements of 3.5.

4.5.7 Graduations.- The accuracy of the scale graduations shall be verified utilizing standard measuring equipment that will produce readings accurate within ten percent of the drawing tolerances. Quality of the printed graduations shall be visually inspected with the aid of a 5 power magnifier to determine conformance with the requirements of 3.6.

4.5.8 Accuracy - "C" and "D" scales.- The test shall be performed in sequence with the test in 4.5.7. Move the center slide until the index of the "C" scale is in coincidence with the index of the "D" scale and clamp the slide and the rule together lightly to prevent movement. Each graduation on the "C" scale shall be checked for alignment with the opposite graduation on the "D" scale, utilizing standard measuring equipment capable of measuring any misalignment of the scale graduations within the tolerance specified. The graduations of the "C" and "D" scales shall conform to the requirements of 3.7.1.

4.5.8.1 Accuracy - ratio "C" and "D" scales.- To determine the ratio of 2/3 for the "C" scale with respect to the "D" scale, utilize the same set-up as specified in 4.5.8. Align the index of the "C" scale opposite the "15" graduation line of the "D" scale and clamp the slide and rule together lightly to prevent movement. At this setting, a ratio of 2/3 shall be read throughout the scale with the opposite graduations in alignment within the tolerance specified in 3.7.1 as required in 3.7.1.1.

4.5.9 Accuracy - site and angle of site readings.- The site and angle of site test for accuracy shall be visually conducted with the aid of an optical device with a magnification of at least 5 power. Position the hairline of the indicator assembly over the specific vertical interval on the "D" scale indicated in Table I for the graphical firing scale. The center slide shall then be moved until the selected range is under the hairline of the indicator assembly. The scale shall be held in this position by applying pressure with the thumb and forefinger of one hand, and with the other hand move the hairline of the indicator assembly to the specified gage point and read the angle of site or site on the "D" scale. The angle of site and site readings shall be accurate within the tolerances specified in Table I.

4.5.10 Accuracy - superelevation readings.- Utilizing an optical device as in 4.5.9, the superelevation scale of the graphical firing scales shall be subjected to a tactile-visual examination to determine its accuracy. The accuracy of the scale shall be checked against Table II. Superimpose the hairline of the indicator assembly over the specified range line, and read the value on the superelevation scale. The readings noted shall be accurate within the tolerances specified in Table II.

4.5.11 Operability.- While stabilized at each temperature extreme specified in 3.3.3 the scale shall move freely when normal thumb pressure is applied to move the slide and the indicator assembly. The slide and the indicator assembly shall be moved several times back and forth to determine conformance with the requirements of 3.8.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with MIL-S-12937. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The scales are special slide rules which are utilized for rapidly calculating firing data pertinent to the weapons systems noted below:

Graphical Firing Scale: XM75E1 - Weapon System: XM28
Graphical Firing Scale: XM76E1 - Weapon System: XM29
Graphical Firing Scale: XM77E1 - Weapon System: XM29

The scales may also be used to solve simple problems in multiplication and division.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of preservation, packaging, and packing.
c. Applicable stock number.

Custodian:

Army - MU

Preparing activity:

Army - MU

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